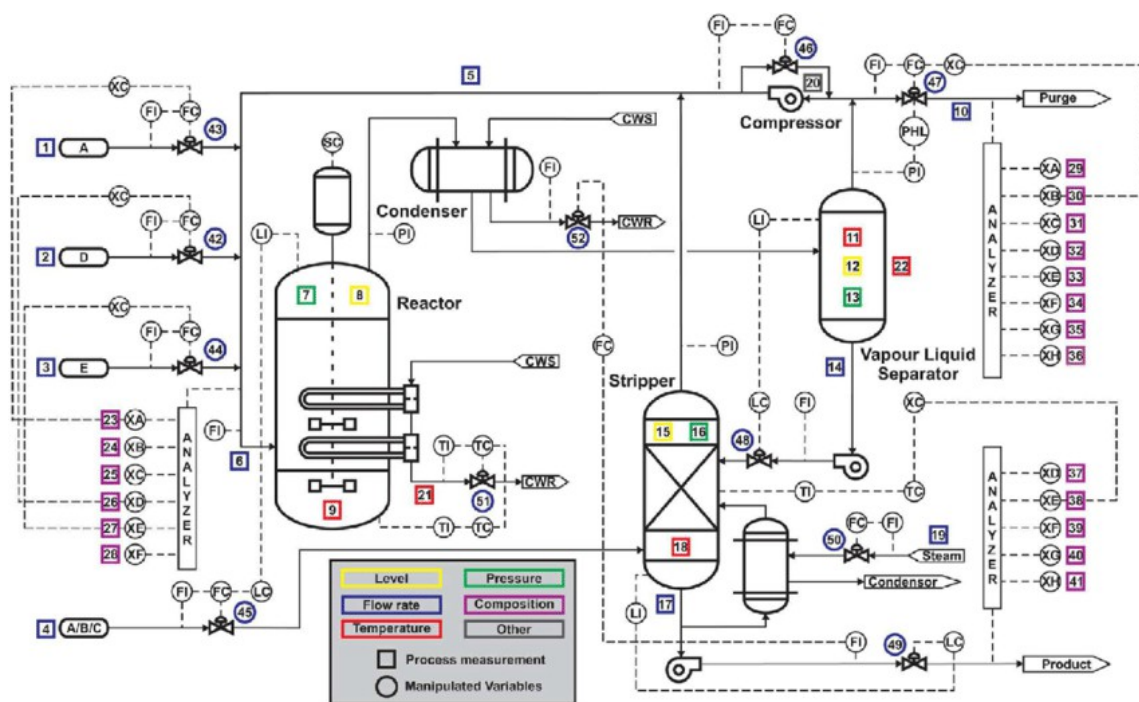


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1. Introduction and Contextualization of the Problem

The Tennessee Eastman Process (TEP) is a mathematical model for simulating a real chemical plant, created and made available by the Eastman Chemical Company in 1993. Since its publication by Downs & Vogel, it has established itself as the global "gold standard" (benchmark) for the development and testing of new technologies in process control, such as PID and MPC, fault detection and diagnosis (FDD), and operational safety. Before the introduction of TEP, academic research was based on overly simplified models, such as level tanks or small furnaces, which did not capture the systemic complexity found in large industrial facilities.



1.1. The Tennessee Eastman Process as an Industrial Benchmark

The importance of TEP lies in its ability to bring real-world complexity into the computational environment, allowing algorithms to be validated in a rigorous setting. It is widely used to evaluate the effectiveness of monitoring and control systems due to its multivariable structure, where adjusting a single variable, such as a separator valve, can dramatically alter the pressure of a reactor in earlier stages of the process. This "everything affects everything" characteristic requires that any control solution be

capable of handling strong couplings and complex interdependencies between plant components.

1.2. Chemical Plant Challenges

The operation of TEP's "virtual factory" poses severe technical challenges, characterized by highly nonlinear behavior, where chemical reactions undergo sudden changes as a function of temperature and pressure. One of the most critical risks is open-loop instability: if the controllers are disabled, the reactor can reach a state of catastrophic failure—a "mathematical explosion"—in less than an hour. In addition, the process deals with exothermic reactions that generate intense heat, requiring constant cooling to prevent overheating and ensure the operational integrity of the system.

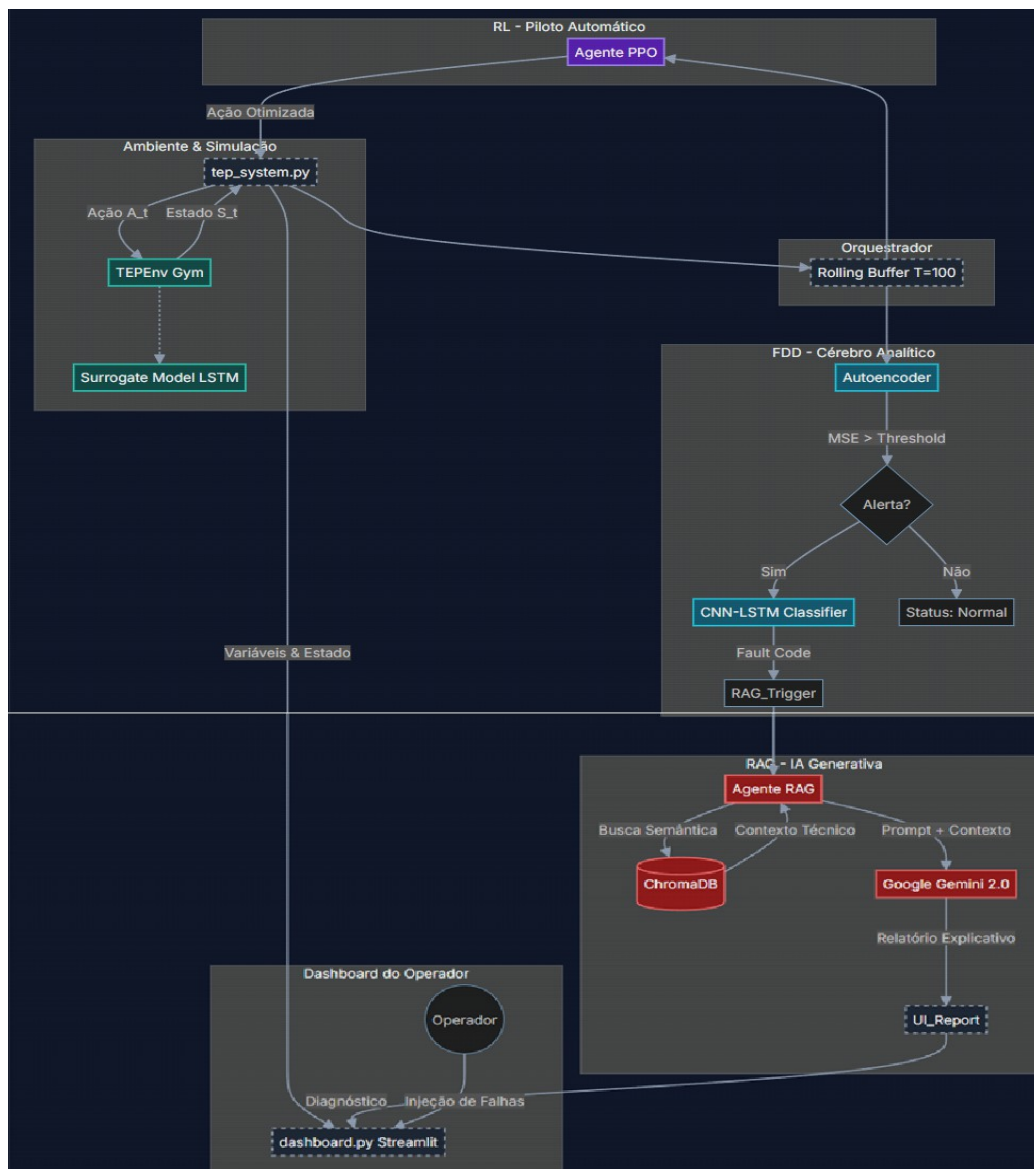
1.3. The TEP-Sentinel Vision

The TEP-Sentinel project was conceived not only as a conventional controller, but as a layer of advanced intelligence positioned above basic process control. Our mission is to use Artificial Intelligence to act proactively: detect subtle deviations before conventional alarms are triggered, diagnose the root cause among the various failures programmed into the simulation, and act automatically, via Reinforcement Learning (RL), to stabilize the plant. In essence, TEP acts as our "flight simulator"; the success of our AI in piloting this unstable and anomaly-ridden chemical plant validates its ability to manage real industrial processes safely and efficiently.

2. System Architecture and Information Flow

In this chapter, we detail the software engineering and integration of the AI models that make up TEP-Sentinel. The system was designed to operate from

modular form, where each component plays a specific role in maintaining stability and the applicability of the industrial process.



Módulos & Componentes

AMBIENTE DIGITAL

Simulação TEP

Digital Twin da planta química Tennessee Eastman. Utiliza um modelo neural (LSTM) treinado em dados históricos para simular a física do processo em tempo real.

`envs/tep_env.py`

DETECÇÃO (FDD)

Monitoramento Inteligente

Rede neural Autoencoder que aprende o comportamento normal da planta. Detecta anomalias sutis (desvios de MSE) e aciona o classificador CNN-LSTM para identificar a causa raiz.

`models/tep_autoencoder.keras`

CONTROLE AUTÔNOMO

Agente PPO

Piloto automático treinado via Reinforcement Learning. Atua sobre 11 válvulas de controle para estabilizar o processo e mitigar falhas, garantindo a segurança operacional.

`models/PPO/best_model1.zip`

IA GENERATIVA

Explainable AI (RAG)

Sistema RAG que traduz códigos de erro em relatórios técnicos. Consulta manuais no ChromaDB e usa o Google Gemini para orientar o operador com ações corretivas.

`rag_agent.py`

FRONTEND

Dashboard Streamlit

Interface de comando e controle. Permite visualizar KPIs em tempo real, injetar falhas para testes e receber diagnósticos da IA.

`dashboard.py`

BACKEND

Orquestrador

Núcleo do sistema ('tep_system.py') que integra o ciclo de sensoriamento, detecção, decisão e explicação (Sense-Think-Act-Explain).

`tep_system.py`

2.1. The Central Orchestrator

The orchestrator acts as the "Central Brain" of the system, responsible for managing the closed-loop operation cycle. Its function is to ensure synchrony between the different layers of the solution:

- **Data Management:** Maintains a rolling window buffer of the last 100 samples to feed the temporal models.
- **Module Integration:** Connects the simulation environment to the detection (FDD), decision (RL), and explanation (RAG) modules.
- **System Stability:** Ensures that models remain loaded in memory for real-time response, eliminating processing latencies.

2.2. The Neural Simulator (Surrogate Model)

Due to the complexity and access restrictions of the original TEP Fortran simulator, we implemented a "Digital Twin" based on a Surrogate Model.

- **LSTM Architecture:** We used a Long Short-Term Memory (LSTM) network trained to predict the next state of the process $S(t+1)$ based on the current state and the control actions applied.
- **Physics Imitation:** The model learned the basic dynamics of the TEP, including the critical correlations between pressure, level, and temperature, with a mean square error (MSE) of ~ 19.2 .
- **Training Infrastructure:** This neural simulator provided the safe and accelerated environment necessary to train the Reinforcement Learning agent without the risk of physical damage to the system.

2.3. "Sense-Think-Act-Explain" Methodology

The TEP-Sentinel architecture follows the Sense-Think-Act-Explain philosophy, integrating three pillars of modern AI into a unified platform to elevate industrial management:

- **Sense:** The environment (TEPEnv) captures the 52 variables monitored on the control panel.
- **Think (FDD Monitoring):** The fault detection module analyzes the data for anomalies and identifies the root cause (Deep Learning).
- **Act (RL Control):** The agent makes autonomous decisions to stabilize the plant and return the variables to their setpoints (Reinforcement Learning).
- **Explain (RAG Explanation):** If a fault is detected, the system consults technical manuals and generates an explanatory report in natural language for the operator (Generative AI).

3. Data Engineering and Processing Pipeline

This section details the statistical and technical rigor applied in data preparation, ensuring that TEP-Sentinel operates on a solid foundation free of biases that could compromise anomaly detection in a real industrial environment.

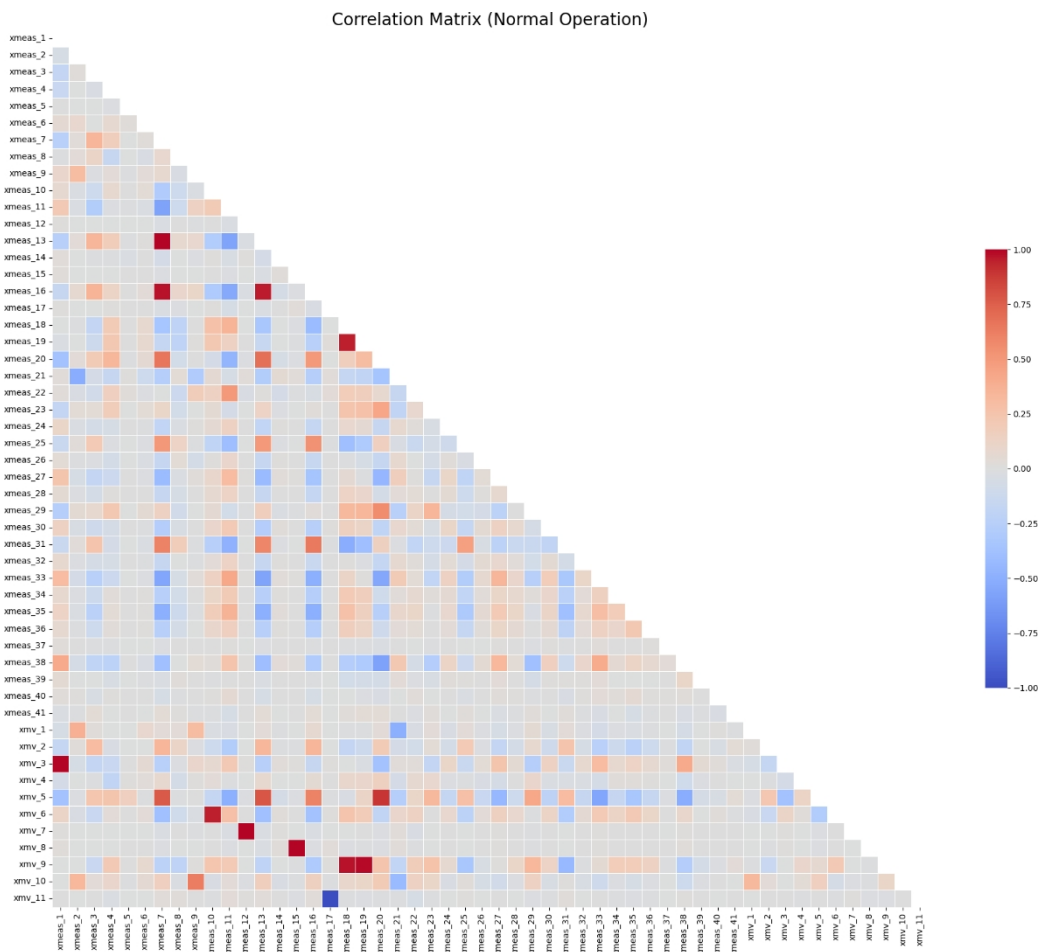
3.1. Ingestion Strategy with Parquet

The original data from the Tennessee Eastman Process simulation is distributed in files in .RData format, native to the R language. To integrate this data into our Deep Learning pipeline in Python, we opted to convert it to Parquet format. This choice was strategic, as Parquet is a columnar storage format that offers superior data compression; while an equivalent CSV file could exceed 2GB, Parquet keeps the data in the hundreds of megabytes. In addition to space efficiency, the format preserves data typing (such as float64 and int) and allows for reading speeds orders of magnitude higher than CSV within the Pandas environment. The implementation used the pyreadr libraries for initial reading and pyarrow as the writing engine.

3.2. Data Leakage Prevention

One of the biggest challenges in fault detection models is avoiding the "leakage" of future or anomalous information into model training.

To mitigate this risk, we use StandardScaler to normalize the data (adjusting the mean to 0 and the standard deviation to 1). The critical point of this step is that the scaler was trained exclusively on data from the FaultFree Training set. By doing so, we ensure that AI models learn the pure statistics of normal operation, without any prior knowledge of fault behavior or the test set, which is essential for a fair assessment of detection capability. The final state of this scaler was persisted in the tep_scaler.pkl file for real-time use in the dashboard.



3.3. Sliding Windows

Industrial data is not static; it has thermal and hydraulic inertia, which means that an isolated sample at time t does not contain enough context for an accurate diagnosis. We implemented a sliding window strategy to capture the temporal dynamics of the process. Each input to the

neural networks consists of a window of $T = 100$ consecutive samples, which represents approximately 3 minutes of operational history.

For the training set, we used a stride of 10 samples, generating overlap between windows as a form of Data Augmentation to increase the robustness of the model. For the test set, we used a stride of 100 (without overlap), ensuring rigorous technical evaluation and memory savings. The final result of this pipeline is tensors compressed into .npz files

.npz files, totaling more than 600,000 sequential samples ready to feed the LSTM and Autoencoders architectures.

4. Intelligent Monitoring and Fault Diagnosis (FDD)

The "brain" of the TEP-Sentinel system consists of a redundant detection and diagnostic structure, acting as a layer of protection between the process and the operator. This phase implements a screening logic where two complementary models work together: the Autoencoder, which acts as a safety "traffic light," and the CNN-LSTM, which takes on the role of a "doctor" to diagnose the specific industrial pathology.

4.1. Unsupervised Detection with Autoencoders

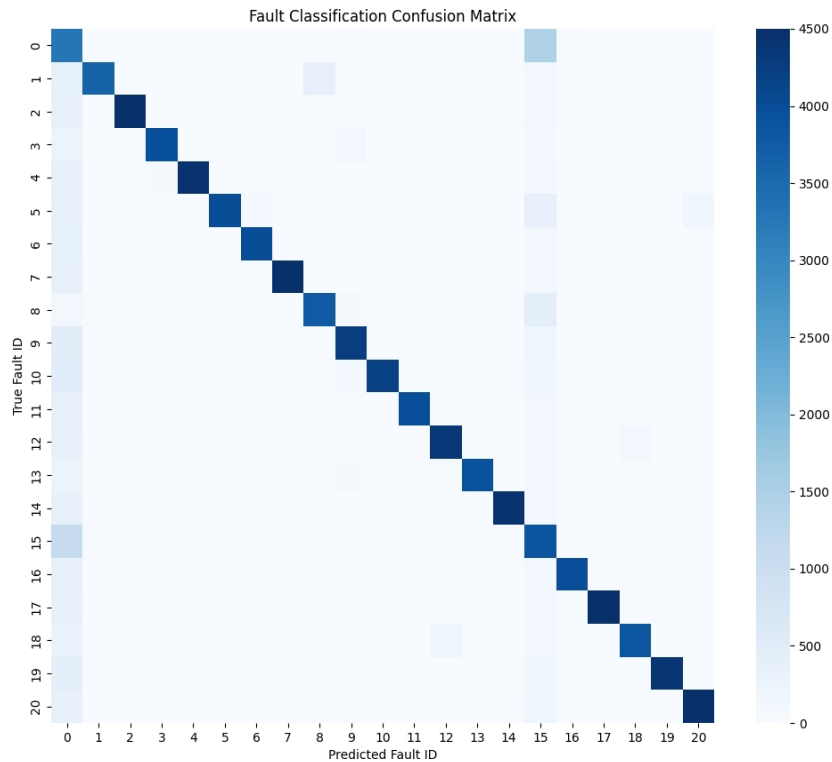
The first line of defense is the Anomaly Detector, an Autoencoder (Encoder-Decoder) type neural network trained under the unsupervised learning paradigm using only normal operating data.

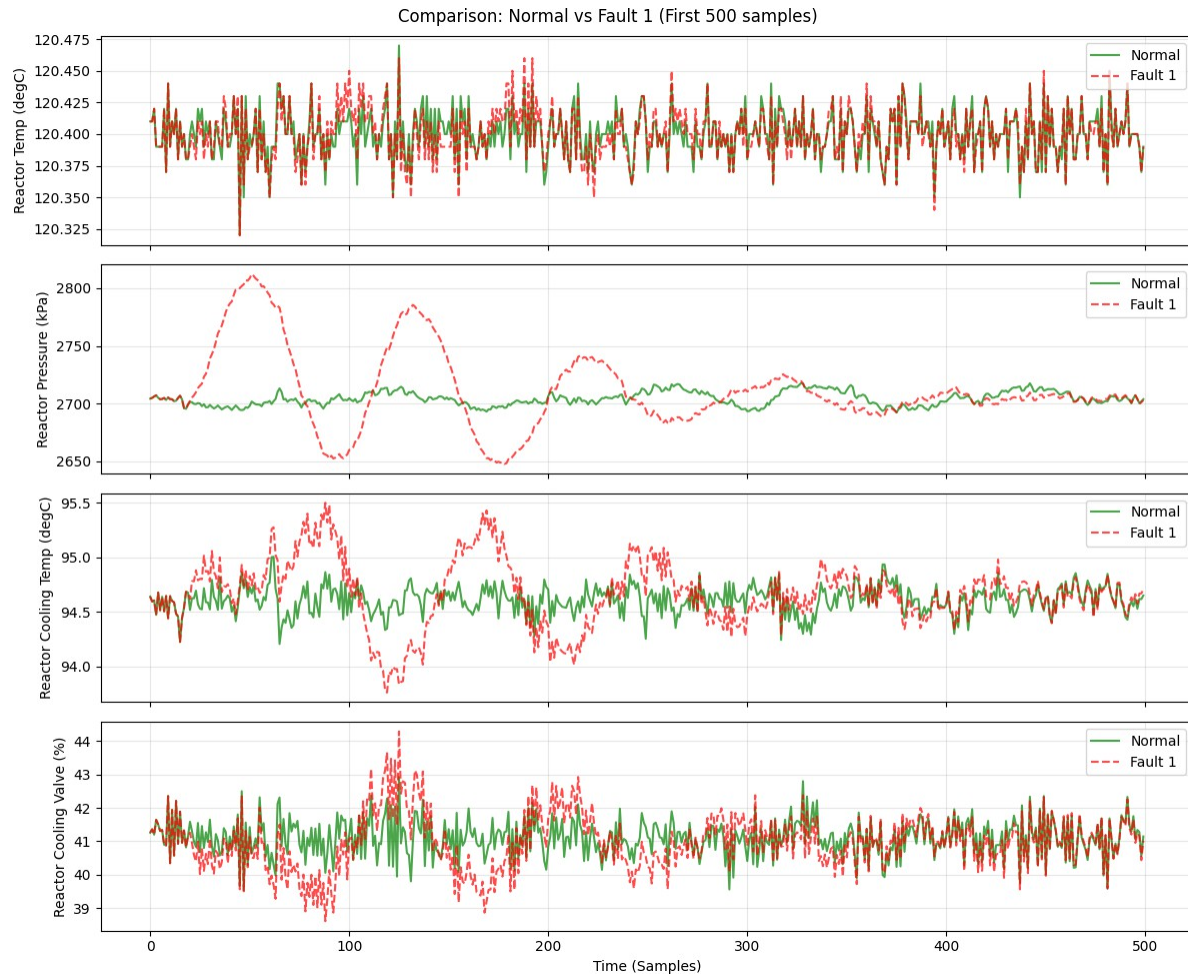
- **Reconstruction Logic:** The model attempts to reconstruct the input window; if the process is in a normal state, the reconstruction error is low.
- **Dynamic Threshold:** We set the alert threshold at the 99th percentile of the Mean Square Error (MSE) of the training.
- **Identification of Unknowns:** If the MSE exceeds this limit (set to 30.7 after calibration), the system marks the status as *FAIL*, being able to detect even anomalies never before seen by the model.

4.2. Root Cause Classification via CNN-LSTM

Once the Autoencoder signals an abnormality, the Fault Diagnostician kicks in to identify the root cause. This model uses an advanced hybrid architecture: CNN extracts spatial features and correlations between the 52 variables, while LSTM interprets the temporal evolution of the deviation.

- **Fault Mapping:** The classifier was trained to distinguish between the 20 programmed faults (IDV 1 to 20).
- **Deviation Analysis:** The model analyzes critical variables such as Reactor Pressure (XMEAS 7) and Reactor Temperature (XMEAS 9), which are highly correlated and serve as "signatures" for various thermal and pressure problems.
- **Practical Example:** In Failure 1 (Change in A/C ratio), the model identifies specific patterns such as the drastic increase in the opening of feed valve A (XMV 3) and the rise in reactor pressure.





4.3. Performance Metrics

The robustness of the FDD system has been statistically validated to ensure reliability in the control room and avoid "alarm fatigue."

Metric	Value Obtained	Operational Impact
FAR (False Alarm Rate)	0.69	Drastic reduction in false alarms, ensuring that the operator trusts the system.
Overall Accuracy	85.55	High reliability in distinguishing between 21 different classes (Normal + 20 Faults).
Average Accuracy	92	The system is extremely accurate when it points to a specific cause.

TPR (True Positive Rate)	88.89	High success rate in detecting Failure 1, capturing most critical deviations.
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The system demonstrates superior performance in critical classes, such as Failure 5 (Temperature), where it achieves 100% accuracy, making it an indispensable tool for reactor safety.

5. Autonomous Control via Reinforcement Learning

In this phase, we implemented the infrastructure for Autonomous Control of the plant, acting as an intelligent "Autopilot" capable of making decisions in fractions of a second to maintain operational stability. Unlike traditional controllers that rely on fixed rules, the Reinforcement Learning (RL) system learns the optimal strategy through continuous interaction with the simulated environment, seeking to minimize deviations and proactively mitigate failures.

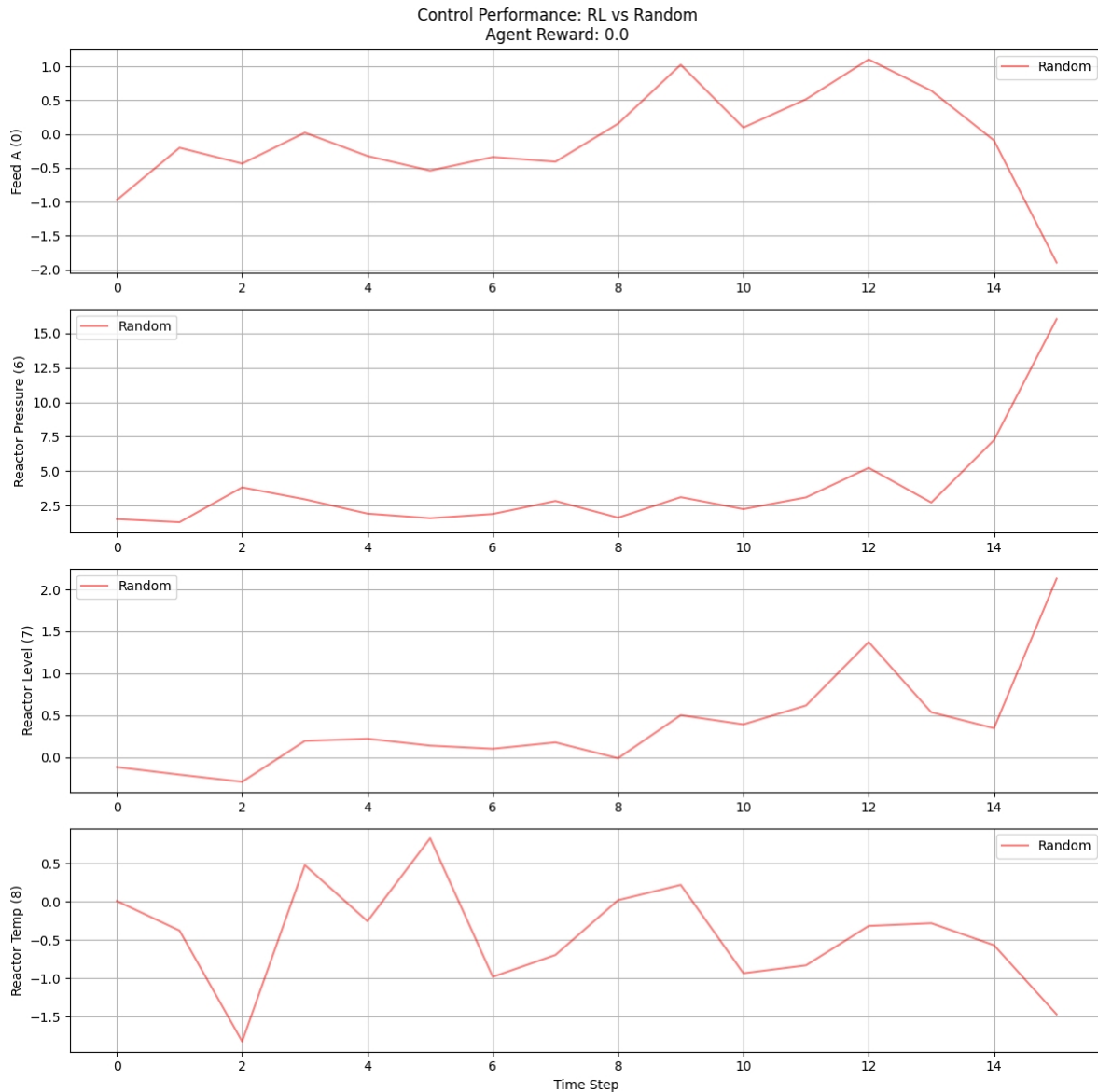
5.1. The PPO (Proximal Policy Optimization) Agent

The "engine" behind our control strategy is the Proximal Policy Optimization (PPO) algorithm, selected for its robustness and efficiency in dealing with continuous action spaces and complex environments. The agent was trained using the stable-baselines3 library to act directly on the 11 manipulated variables (XMV), which represent the openings of the plant's feed, recycle, and purge valves. The central objective of PPO is to keep the measurement variables (XMEAS) as close as possible to their ideal setpoints (zero value after normalization), ensuring that the production of saleable products (G and H) occurs in a stable manner even under external disturbances.

5.2. Safety Layer

One of the biggest challenges in applying AI to critical industrial processes is ensuring that the agent does not take catastrophic actions during the learning process. To address this, we have implemented a Safety Layer.

(Safety Layer) hard-coded directly into the `envs/tep_env.py` environment. This mechanism continuously monitors the plant's status and applies severe penalties to the agent if it suggests actions that lead to critical deviations, defined as variations greater than 10σ (ten standard deviations) from the stable state. This physical barrier prevents reactor runaway (mathematical explosion) and ensures that, even in pursuit of efficiency, the system never violates physical safety constraints.



5.3. Real-Time Optimization and Inference

The effectiveness of an industrial controller depends on its responsiveness. To enable real-time training and operation, we use a

Surrogate Model (LSTM) that emulates the physics of TEP with high fidelity. We implemented JIT (Just-In-Time) compilation through the `@tf.function` decorator in the simulation environment, which resulted in a 50x acceleration in processing speed, raising the inference rate from ~2 steps per second to approximately 100 steps per second. This performance gain allows the system to process data time windows, execute diagnostic models, and take control actions almost instantaneously, a critical requirement for managing the rapid pressure and temperature dynamics of the reactor.

6. Explainable AI (RAG) and Operator Interface

In this section, we describe how TEP-Sentinel translates the complexity of industrial data into actionable information for humans, using generative AI technologies and interactive interfaces.

6.1. RAG Module with Gemini 2.0

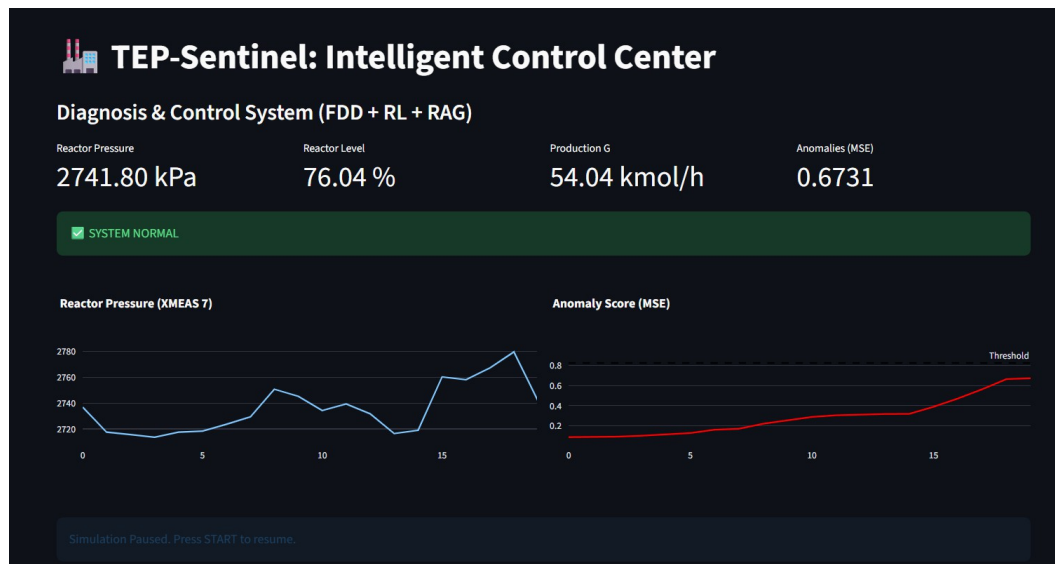
The system uses the RAG (Retrieval-Augmented Generation) technique to provide explainable diagnostics. The process begins when the detection module identifies a fault and sends the corresponding code (e.g., IDV 1) to the RAG Agent.

- **Semantic Search:** The agent queries a vector database (ChromaDB), which contains indexed technical manuals and plant documentation.
- **Natural Language Processing:** Using the Google Gemini 2.0 Flash model, the system cross-references the retrieved data with the context of the fault to generate a technical report in natural language.
- **Accuracy:** This approach eliminates the need for the operator to memorize error codes, transforming a mathematical alert into a clear explanation of the root cause.

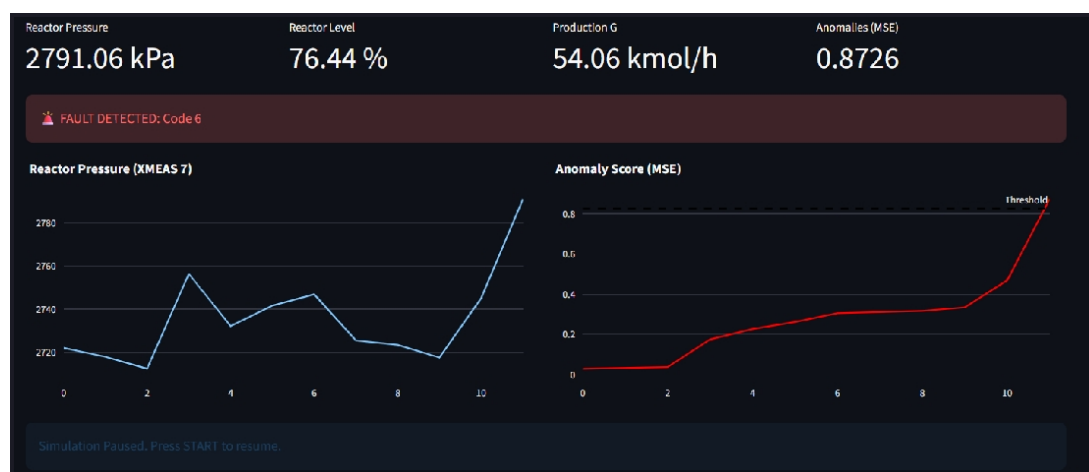
6.2. Streamlit Dashboard

The interface of command of the project was developed in Streamlit (dashboard.py), prioritizing real-time visualization and ease of use.

- **Simulation Control:** The operator can start, stop, or reset the simulation directly from the dashboard.
- **KPI Visualization:** The dashboard displays dynamic graphs of critical variables, such as Reactor Pressure and Level, in addition to the anomaly score (MSE) calculated by the Autoencoder.
- **Fault Injection:** For testing and robustness validation purposes, the interface allows specific anomalies to be injected and the immediate response of the systems to be observed detection detection and



control systems.



6.3. Decision Support (Co-pilot)

TEP-Sentinel acts as a "Virtual Senior Engineer," helping to democratize technical knowledge within the plant.

- **MTTR Reduction:** By suggesting immediate corrective actions based on manuals (e.g., "check valve X," "reduce flow Y"), the system dramatically reduces Mean Time to Repair.
- **Emergency Support:** In high-pressure scenarios, AI provides decision support that is less dependent on the operator's individual experience, consulting terabytes of manuals in seconds.

EXAMPLE TO FOLLOW:

Fault Analysis and Recommendations

1. Fault Identification

Based on the Alert Code IDV(6), the technical name of the fault is Temp. Failure (Failure related to Temperature). This is inferred from the "Highlights by Class" table in the Fault Diagnosis section, where Class 5 corresponds to "Temp. Failure."

2. Root Cause Explanation

While the provided context does not specify which temperature is failing, the high MSE error (2.2333) indicates a significant deviation from normal operating conditions. Given the strong correlation between Reactor Pressure and Temperature, a temperature fault could be related to issues within the reactor itself, or in systems affecting the reactor temperature, such as feed preheating or cooling systems.

3. Recommended Actions for the Operator

Given the "Temp. Failure" alert and the high MSE error, the operator should take the following actions:

Immediate Actions:

Verify Reactor Temperature (XMEAS 7): Confirm the reactor temperature reading and compare it to the setpoint. A significant deviation warrants immediate investigation.

Check Reactor Pressure (XMEAS 7): As temperature and pressure are highly correlated, verify the reactor pressure reading. A simultaneous deviation in pressure and temperature strongly suggests a reactor-related issue.

Acknowledge and Log the Alarm: Ensure the alarm is acknowledged in the control system and a detailed log entry is created, documenting the time, alarm details, and initial observations.

Investigative Actions:

Examine Feed Temperatures: Check the temperatures of the incoming feed streams (A, C, D, E) to ensure they are within the expected ranges. A deviation in feed temperature can directly impact the reactor temperature.

Inspect Cooling System: Verify the functionality of the reactor cooling system. Check coolant flow rates, coolant temperature, and the status of any associated pumps or valves.

Review Reactor Heating System: If the reactor uses a heating system (e.g., steam tracing), check the steam supply pressure and flow rate.

Analyze Recent Process Trends: Review recent trends of relevant process variables (temperatures, pressures, flow rates) to identify any patterns or anomalies that may have preceded the alarm.

Check for Fouling: If the temperature deviation is gradual, consider the possibility of fouling on heat transfer surfaces within the reactor or associated heat exchangers.

Valve Checks:

Cooling Water Valve: Verify the position and operation of the cooling water valve controlling the reactor temperature.

Steam Valve (if applicable): If steam is used for heating, check the steam valve position and ensure it is responding correctly.

Setpoint Adjustments:

Avoid immediate setpoint adjustments: Do not attempt to correct the temperature by drastically changing setpoints without first understanding the root cause. This could mask the underlying problem or potentially exacerbate the situation. Only make small, controlled adjustments if necessary, and carefully monitor the response.

Important Considerations:

Safety First: Prioritize safety in all actions. If the temperature deviation is significant and poses a safety risk, initiate emergency shutdown procedures as necessary.

Consult with Experts: If the root cause is not immediately apparent, consult with process engineers or other experts for further guidance.

Document Everything: Maintain a detailed record of all actions taken, observations made, and any changes to process parameters. This documentation will be invaluable for future troubleshooting and analysis.

7. Industrial Feasibility and Conclusion

The conclusion of this report focuses on the transition of the project from a simulation environment to a real application in refineries or chemical plants.

7.1. OT/IT Convergence

For industrial implementation (Industry 4.0), the TEP-Sentinel architecture must evolve to support convergence between operational technology (OT) and information technology (IT).

- Physical Layer (OT): Field sensors connected to programmable logic controllers (PLCs) via protocols such as Modbus or OPC-UA.

- **Edge Computing:** Locally installed industrial gateways must run the detection module (Autoencoder) to ensure zero latency in the detection of critical anomalies, without total dependence on the cloud.

7.2. The Industrial AI Tripod

The project successfully demonstrates the effectiveness of orchestrating three fundamental pillars of modern AI in the manufacturing sector:

- **Monitoring** **Unsupervised:** Autoencoders that detect
the unknown.
- **Intelligent Control:** Reinforcement Learning (RL) for optimization beyond human capability.
- **Explainability:** Generative AI (RAG) that translates complex mathematics for the operator.

7.3. Economic Impact

The implementation of this ecosystem generates tangible business value by transforming raw data into strategic decisions.

- **Efficiency** **Operational:** Reduction in downtime and waste
waste of raw materials through early detection and adaptive control.
- **Knowledge Management:** The system mitigates "brain drain" by digitizing and making the tacit knowledge of experienced operators searchable.

TEP-Sentinel proves that the true industrial revolution does not lie in isolated tools, but in collaboration between humans and AI, where machines take care of operational complexity and humans focus on strategy (Industry 5.0).

